AIF 2026 · Phase 2 · Week 4

Statistical Machine Learning: Linear Models

Weekly Assignment

Field	Detail
Programme	AI Fellowship 2026
Phase	Phase 2 Foundational Statistical Learning
Week	Week 4
Topic	Statistical Machine Learning: Linear Models
Dataset	Telco Customer Churn (Kaggle)
Submission	Single Jupyter Notebook (.ipynb), fully executed

Context

You are a Data Scientist at a telecom company. The business is losing customers to churn and wants to take action. You have been handed historical customer data and one week to deliver a production-ready recommendation.

That means understanding the problem deeply, running experiments across multiple models, evaluating your results honestly, and making a justified decision about what to ship. There is no prescribed model. You decide and you must defend every decision with evidence from your experiments.

> This is not a tutorial. There are no step-by-step instructions. You are expected to make decisions, justify them, and take ownership of your results. Bring your questions, your failures, and your findings to the session.

Dataset

Source: Telco Customer Churn
https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Size: 7,043 rows · 21 columns · Single CSV file

Features

Category	Columns
Demographics	gender, SeniorCitizen, Partner, Dependents

Services	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Account Info	Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, tenure
Target	Churn (Yes / No)

Known Issues

These are not pre-cleaned for you. Handling them is part of the assignment.

- TotalCharges contains whitespace strings (' ') where values are missing. You need to detect this, decide how to handle it, and justify your decision.
- The dataset is imbalanced: approximately 27% of customers churned. This affects model training, metric choice, and threshold decisions.
- Features are a mix of numeric and categorical types. Encoding decisions are yours to make.

Tasks

The assignment is structured into five tasks. Each task builds on the previous one. Do not skip ahead.

Task 1 Understand the Problem First

Before writing any modelling code, you must understand what you are actually solving. This task is about formulating the problem and understanding the data.

1. Formulate the ML problem formally. What is X (your feature space)? What is y (your target variable)? What probability distribution naturally models the target, and what does that tell you about the right loss function?
2. What assumptions are you making about the data-generating process? List at least three. For each assumption, describe what would happen to your model if that assumption were violated.
3. Identify sources of uncertainty in this dataset. Where is the data noisy, biased, or incomplete? Be specific, name the columns and describe the issue.
4. Profile the distributions of MonthlyCharges, tenure, and TotalCharges. What shape does each distribution follow? Are there impossible or suspicious values? Decide how to handle them and justify every decision.
5. The dataset has approximately 27% churn. Build a naive baseline: a model that always predicts the majority class. What accuracy does it achieve? Why is this accuracy misleading and why is this model dangerous to rely on?

Task 2 Classification Experiment Who Will Churn?

This is an open experiment. Your goal is to find the best linear classifier for predicting churn. You are not told which model to use. You decide, you experiment, and you justify your choice.

> Candidates to explore: Logistic Regression, Ridge Classifier, SGD Classifier. You are not required to use all of them but you must justify what you tried and why you chose what you chose.

1. For each model you try: train it, evaluate it on the validation set, and record your results in a comparison table.
2. Evaluate using metrics appropriate for an imbalanced dataset. Think carefully before reporting accuracy. Which metrics actually tell you whether the model is useful to the business? Your comparison table must include: Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, and Log Loss.
3. Plot ROC and Precision-Recall curves for your best model. The retention team has a limited budget: they can only call 200 customers per week. At what threshold would you deploy your model given this constraint? Justify your answer.

4. Inspect the coefficients of your chosen model. Which features drive churn the most? Do the signs and magnitudes make business sense? If any coefficient surprises you, investigate why.
5. Compare Batch Gradient Descent (LogisticRegression) and Stochastic Gradient Descent (SGDClassifier). Do they converge to the same solution? Which is faster? When would you prefer one over the other?

Task 3 Regression Experiment How Long Will They Stay? What Are They Worth?

Binary churn prediction tells you who might leave. Regression tells you how long they will stay and how much they are worth. This task asks you to build regression models for at least one of the following targets.

Choose at least one regression target:

- (a) Tenure as survival time how many months until a customer churns
- (b) Predicted churn probability score using the output of your classification model as a continuous target
- (c) Customer Lifetime Value (CLV) = MonthlyCharges × predicted tenure

1. This is again an open experiment. Candidates to explore: Linear Regression, Ridge, Lasso, Elastic Net. Try at least two. Compare results in a table and justify your final choice.
2. Evaluate using MAE, RMSE, and R^2. Do not just report the numbers and interpret each one. What does an R^2 of 0.55 actually mean in this context?
3. Plot residuals against predicted values. Is there a pattern? What does that tell you about model fit or assumption violations?
4. Apply Ridge, Lasso, and Elastic Net regularization. Compare coefficients across different regularization strengths. Plot the Lasso regularization path. Which features survive at high regularization? Explain geometrically why L1 produces sparse solutions and L2 does not.
5. What does CLV enable the business to do that binary churn prediction alone cannot?

Task 4 Evaluation Integrity

A model with inflated metrics is worse than a model with honest ones. This task is about verifying that your evaluation is trustworthy.

1. Split your data correctly into training, validation, and test sets. Justify your split strategy. Check explicitly for leakage between splits.
2. Run k-fold cross-validation on your best classification model. How does CV performance compare to holdout performance? Why might they differ?

3. Plot learning curves for your best model. Is it underfitting, overfitting, or well-generalised? What is the correct intervention in each case?
4. Deliberately introduce a leakage feature into your pipeline. Observe what happens to your metrics. Document the before and after. Explain why this is catastrophic in a production system.

Task 5 Production Decision

You have run your experiments. Now you must commit to a decision and defend it.

1. Based on all your experiments, which classification model do you deploy? Which regression model? Write a model card (approximately half a page) covering:
    - Chosen model and key hyperparameters
    - Key metrics on the held-out test set
    - Deployment threshold and justification
    - Known limitations of the model
    - What could go wrong in production
    - What you would monitor after deployment
2. Are linear models sufficient for this problem, or do you think a more complex model is needed? What specific evidence from your experiments supports that view?

Submission Requirements

What to Submit

One single Jupyter Notebook (.ipynb) file. Nothing else.

The Notebook Must:

- Be fully executed with all outputs visible. Do not submit a notebook with empty output cells.
- Contain clean, well-structured, and well-commented code. Code that cannot be understood will not be graded.
- Include written explanations alongside every result. Numbers without interpretation are not answers.
- Address all five tasks in order. Each task must be clearly labelled with a markdown header.
- Include a model card as a markdown cell in Task 5.

Code Quality Standards

Standard	Expectation
Structure	Sections clearly separated with markdown headers matching the five tasks
Comments	Every non-trivial code block has a comment explaining what it does and why
Output	All cells executed top-to-bottom with outputs visible
Variable names	Descriptive variable names not x1, x2, df2
Plots	All plots have titles, axis labels, and legends where applicable
Written reasoning	Every decision (model choice, metric choice, threshold, handling of nulls) is explained in prose

> *If you cannot explain a piece of code or a decision in your notebook, you have not solved the problem. You have outsourced it. You will be asked to explain your work in the session.*

Evaluation Criteria

Component	Weight	What We Look For

Problem Formulation	15%	Formal problem statement, assumptions stated, uncertainty identified
Classification Experiment	25%	Experiments run, comparison table complete, threshold justified
Regression Experiment	20%	Regression target derived, models compared, regularization explored
Evaluation Integrity	20%	Leakage demo, learning curves, CV vs holdout comparison
Production Decision	20%	Model card complete, decision justified with evidence

Honour Code

AI tools and LLMs are allowed for learning, brainstorming, debugging, and prototyping. They are tools to assist your thinking, not to replace it.

- You must be able to explain every line of code in your notebook.
- You must be able to explain every decision, every metric, and every number.
- If you cannot explain it in the session, it will not count toward your grade.
- Do not submit work you cannot defend. You will just look stupid.

> The point of this assignment is not to get good numbers. It is to understand what those numbers mean, make a decision you can defend, and communicate it clearly. A student who achieves PR-AUC of 0.55 and can fully explain every decision will be valued more than one who achieves 0.72 and cannot explain why.

Resources

Dataset

- Telco Customer Churn https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Reading

- ISLP (James et al.) Ch. 3: Linear Regression, Ch. 4: Classification, Ch. 6: Regularization https://www.statlearning.com
- Applied ML and AI for Engineers Jeff Prosise

Videos

- Fusemachine Microdegree Machine Learning

NotebookLLM:

- Research based on the source books and links
    - https://notebooklm.google.com/notebook/acdf599d-c1d6-4b39-b967-d8fca56a98c2